Global wellposedness for the Maxwell-Klein-Gordon equation below the energy norm

Terence Tao · 12 October 2009

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Mark Keel, Tristan Roy, and I have just uploaded to the arXiv the paper "Global well-posedness for the Maxwell-Klein-Gordon equation below the energy norm", submitted to Discrete and Continuous Dynamical Systems. This project started about eight years ago, and was in fact a partial result was essentially finished by 2002, but managed to get put on the backburner for a while due to many other priorities. Anyway, this paper applies the I-method to the Maxwell-Klein-Gordon system of equations in the Coulomb gauge (a simplified model for the hyperbolic Yang-Mills equations) in three spatial dimensions. Previously to this paper, it was known that the Cauchy problem was globally well-posed in the energy norm (which is essentially $H^1(\mathbb{R}^3)$) and locally well-posed in $H^s(\mathbb{R}^3)$ for $s > 1/2$, with this condition being essentially best possible except for the endpoint. Here we manage to lower the regularity threshold for global wellposedness to $s > \sqrt{3}/2 \approx 0.866$. (The partial result alluded to earlier was for $s > 7/8 = 0.875$; at one point we had announced an improvement to $5/6 \approx 0.833$, but the argument turned out to be flawed.) This is part of what is now quite a large family of such "global well-posedness below the energy norm" results, but there are some notable technical features here which were not present in earlier works. Firstly, we can show that there is no smoothing effect in the nonlinearity, ruling out use of the Fourier truncation method. Secondly, due to our use of rescaling, supercritical quantities such as the L^2 norm are not under control, which necessitates some unusual treatment of the low frequency portions of the scalar and vector fields. Namely, they are estimated in L^p spaces rather than L^2 ones. This complicates a number of tasks, ranging from controlling the elliptic theory, to understanding the coercive nature of the Hamiltonian, to establishing the nonlinear commutator estimates underlying the almost conservation law.

Comments (4)



Aaron Sterling · 14 October 2009

I apologize for the off-topic comment, but I'm not sure where to ask this.
I'm part of a seminar that has started to review Additive Combinatorics, and there's a fact known as the "Popularity Principle." (Problem 1.1.4, page 5.) We can't figure out why it bears this name, and Google doesn't seem to help. Could someone enlighten me, please?



Terence Tao · 14 October 2009

My blog page for the book,

> Additive combinatorics

is probably the most appropriate place for this type of question.
In any case, the name derives from Gowers' first paper on his proof of Szemeredi's theorem, in which he starts with a collection of differences a-b (with a in A and b in B) in a small set, and extracts out the "popular" differences – those differences d with many representations in the form a-b. The point is that most differences a-b (counting multiplicity tend to be popular, simply because unpopular differences d don't arise from too many pairs (a,b).



Aaron Sterling · 14 October 2009

Thank you. I'll post in the right place next time if other things come up. :-)



A semana nos arXivs... « Ars Physica · 17 October 2009

[...] Global well-posedness of the Maxwell-Klein-Gordon equation below the energy norm. (arXiv:0910.1850v1 [math.AP]), com direito a comentários do próprio autor: Global wellposedness for the Maxwell-Klein-Gordon equation below the energy norm [...]